

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – C Code Output Prediction

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	Assessment:	Assessment Tool 1 – C Code Output Prediction
Weightage:	35% of CIA	Total Marks:	100
CO1 Statement:	Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.		
Student Name:	Jaswanth.S	Reg. No.:	192525214
Section / Batch:	DS Programming	Date:	10/07/2026

Code of Conduct

I Jaswanth.S / 192525214 (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: **Jaswanth.S**

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Linked data structure	Basic data structure	1 - 2	20
Primitive and Non-Primitive Data Structure - Linear and Non-Linear Data Structure	Classifications of Linear data structure	3 - 4	20
Search Data Structure	Linear Search and Binary Search	5 - 6	20
Recursion, Performance analysis	Recursion	7 - 8	20
Array Data Structures	Implementations Using Array data structure	9 - 10	20
		GRAND TOTAL:	100 Marks

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CO1_AT1_C Code Output Prediction — Answers

Q1 C Code – Array Traversal (10 marks)

```
#include <stdio.h>
int main() {
int arr[] = {5, 10, 15, 20};
int i, sum = 0;
for(i = 0; i < 4; i++)
sum += arr[i];
printf("%d", sum);
return 0;
}
```

Answer: 50

Screenshot:

The screenshot displays the SIMATS Online Programming Lab interface. At the top, there's a browser window showing the URL: `simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=330`. Below the browser, the SIMATS Online Programming Lab header is visible, with the SIMATS ENGINEERING logo and the text "DS PROGRAMMING". The main content area is titled "CO1_AT1_C CODE OUTPUT PREDICTION". On the left, there's a "QUESTIONS" sidebar with a list of questions, each marked with a green checkmark. The first question, "Q1 C Code – Array Traversal", is selected. The main area shows the C code for Q1, which is the same code as in the previous block. Below the code, there's a "Your Answer" section with a text input field containing the number "50". At the bottom of the answer section, there's a blue button labeled "Save Answer".

Q2 C Code – Linear Search (10 marks)

```
#include <stdio.h>
int main() {
int arr[] = {2,4,6,8,10};
int key = 8;
int i;
for(i=0;i<5;i++){
if(arr[i]==key){
printf("%d",i);
break;
}}}
```

Answer: 3

Screenshot:

The screenshot displays the SIMATS Online Programming Lab interface. At the top, there's a navigation bar with the SIMATS ENGINEERING logo and the text "SIMATS Online Programming Lab DS PROGRAMMING". Below this, the page title is "CO1_AT1_C CODE OUTPUT PREDICTION". On the left, a sidebar lists eight questions (Q1 to Q8) with their respective topics and marks. Q2, "C Code – Linear Search", is selected. The main area shows the C code for Q2, which is identical to the code provided in the first block. Below the code, there's a "Your Answer" section where the number "3" has been entered. A "Save Answer" button is located at the bottom of the answer section.

QUESTIONS

- Q1 C Code – Array Traversal 10 marks
- Q2 C Code – Linear Search 10 marks
- Q3 C Code – Binary Search Mid 10 marks
- Q4 C Code – Pointer with Array 10 marks
- Q5 C Code – Array Update 10 marks
- Q6 C Code - Linked List Node 10 marks
- Q7 C Code - Time Complexity Loop 10 marks
- Q8

Q2 C Code – Linear Search

```
#include <stdio.h>
int main() {
int arr[] = {2,4,6,8,10};
int key = 8;
int i;
for(i=0;i<5;i++){
if(arr[i]==key){
printf("%d",i);
break;
}}}
```

Your Answer

3

Save Answer

Q3 C Code – Binary Search Mid (10 marks)

```
#include<stdio.h>
int main(){
int low=0,high=7;
int mid;
mid=(low+high)/2;
printf("%d",mid);
}
```

Answer: 3

Screenshot:

The screenshot displays the SIMATS Online Programming Lab interface. At the top, there's a navigation bar with the SIMATS Engineering logo and the text "SIMATS Online Programming Lab" and "DS PROGRAMMING". Below this, the page title is "CO1_AT1_C CODE OUTPUT PREDICTION". On the left, a sidebar lists eight questions (Q1 to Q8) with their respective topics and marks. Q3, "C Code – Binary Search Mid", is selected. The main area shows the C code snippet for Q3, followed by a "Your Answer" input field where the number "3" has been entered. A "Save Answer" button is located below the input field.

QUESTIONS

- Q1 C Code – Array Traversal 10 marks
- Q2 C Code – Linear Search 10 marks
- Q3 C Code – Binary Search Mid 10 marks
- Q4 C Code – Pointer with Array 10 marks
- Q5 C Code – Array Update 10 marks
- Q6 C Code - Linked List Node 10 marks
- Q7 C Code - Time Complexity Loop 10 marks
- Q8

Q3 C Code – Binary Search Mid

```
#include<stdio.h>
int main(){
int low=0,high=7;
int mid;
mid=(low+high)/2;
printf("%d",mid);
}
```

Your Answer

3

Save Answer

Q4 C Code – Pointer with Array (10 marks)

```
#include<stdio.h>
int main(){
int arr[]={10,20,30};
int *p=arr;
printf("%d", *(p+2));
}
```

Answer: **30**

Screenshot:

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```
#include<stdio.h>
int main(){
int arr[]={10,20,30};
int *p=arr;
printf("%d", *(p+2));
}
```

 Below the code, the "Your Answer" section shows the input "30". A "Save Answer" button is located at the bottom of the answer section.

Q5 C Code – Array Update (10 marks)

```
#include<stdio.h>
int main(){
int arr[]={1,2,3};
arr[1]=arr[0]+arr[2];
printf("%d",arr[1]);
}
```

Answer: 4

Screenshot:

The screenshot shows a web browser window with the SIMATS Online Lab interface. The browser's address bar shows the URL: `simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=330`. The page header includes the SIMATS ENGINEERING logo and the text "SIMATS Online Programming Lab" with a "DS PROGRAMMING" tag. The main content area is titled "CO1_AT1_C CODE OUTPUT PREDICTION". On the left, a list of questions (Q1 to Q8) is shown, each with a green checkmark indicating it has been completed. The question Q5, "C Code – Array Update", is selected. The code for Q5 is displayed in a text area:

```
#include<stdio.h>
int main(){
int arr[]={1,2,3};
arr[1]=arr[0]+arr[2];
printf("%d",arr[1]);
}
```

 Below the code, the "Your Answer" section shows the number "4". A blue button labeled "Save Answer" is located at the bottom of the answer section.

Q6 C Code – Linked List Node (10 marks)

```
#include<stdio.h>
struct Node{
int data;
struct Node *next;
};
int main(){
struct Node n1,n2;
n1.data=10;
n2.data=20;
n1.next=&n2;
n2.next=NULL;
printf("%d",n1.next->data);
}
```

Answer: 20

Screenshot:

The screenshot shows a web browser window with the SIMATS Online Programming Lab interface. The browser's address bar shows the URL: `simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=330`. The page header includes the SIMATS ENGINEERING logo and the text "SIMATS Online Programming Lab" and "DS PROGRAMMING".

The main content area is titled "CO1_AT1_C CODE OUTPUT PREDICTION". It features a sidebar on the left with a list of questions (Q1 to Q8) and their respective topics and marks. The question "Q6 C Code - Linked List Node" is highlighted in blue. The main area displays the C code for Q6, and the "Your Answer" section shows the predicted output "20".

QUESTIONS

- Q1 C Code – Array Traversal 10 marks
- Q2 C Code – Linear Search 10 marks
- Q3 C Code – Binary Search Mid 10 marks
- Q4 C Code – Pointer with Array 10 marks
- Q5 C Code – Array Update 10 marks
- Q6 C Code - Linked List Node 10 marks**
- Q7 C Code - Time Complexity Loop 10 marks
- Q8

Q6 C Code - Linked List Node

```
#include<stdio.h>
struct Node{
int data;
struct Node *next;
};
int main(){
struct Node n1,n2;
n1.data=10;
n2.data=20;
n1.next=&n2;
n2.next=NULL;
printf("%d",n1.next->data);
}
```

Your Answer

20

Q7 C Code – Time Complexity Loop (10 marks)

```
for(i=0;i<n;i++)  
printf("**");  
The complexity is
```

Answer: $O(n)$

Screenshot:

The screenshot shows a web browser window with the SIMATS Online Lab interface. The browser's address bar shows the URL: `simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=330`. The page header includes the SIMATS ENGINEERING logo and the text "SIMATS Online Programming Lab" with a "DS PROGRAMMING" tag. The main content area is titled "CO1_AT1_C CODE OUTPUT PREDICTION". On the left, a sidebar lists eight questions (Q1 to Q8) with their respective topics and marks. Question Q7, "C Code - Time Complexity Loop", is selected. The main area displays the C code for Q7:

```
for(i=0;i<n;i++)  
printf("**");  
The complexity is
```

 Below the code, the "Your Answer" section shows the input `O(n)`. A "Save Answer" button is located at the bottom of the answer section.

QUESTIONS

- Q1**
C Code – Array Traversal
10 marks
- Q2**
C Code – Linear Search
10 marks
- Q3**
C Code – Binary Search Mid
10 marks
- Q4**
C Code – Pointer with Array
10 marks
- Q5**
C Code – Array Update
10 marks
- Q6**
C Code - Linked List Node
10 marks
- Q7**
C Code - Time Complexity Loop
10 marks
- Q8**

Q7 C Code - Time Complexity Loop

```
for(i=0;i<n;i++)  
printf("**");  
The complexity is
```

Your Answer

O(n)

Save Answer

Q8 C Code – Nested Loop (10 marks)

```
for(i=0;i<n;i++)  
for(j=0;j<n;j++)  
printf("**");
```

Answer: $O(n^2)$

Screenshot:

The screenshot shows a web browser window with the SIMATS Online Lab interface. The browser's address bar shows the URL: `simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=330`. The page header includes the SIMATS ENGINEERING logo and the text "SIMATS Online Programming Lab" with a "DS PROGRAMMING" tag. The main content area is titled "CO1_AT1_C CODE OUTPUT PREDICTION". On the left, a list of questions (Q1 to Q8) is shown, each with a green checkmark and a description of the problem. The selected question, Q8, is "C Code - Nested Loop" and is displayed in a larger box on the right. The code for Q8 is:

```
for(i=0;i<n;i++)  
for(j=0;j<n;j++)  
printf("**");
```

 Below the code, the text "Your Answer" is followed by a text input field containing the answer "O(n2)". A blue button labeled "Save Answer" is located below the input field.

QUESTIONS

- Q1**
C Code – Array Traversal
10 marks
- Q2**
C Code – Linear Search
10 marks
- Q3**
C Code – Binary Search Mid
10 marks
- Q4**
C Code – Pointer with Array
10 marks
- Q5**
C Code – Array Update
10 marks
- Q6**
C Code - Linked List Node
10 marks
- Q7**
C Code - Time Complexity Loop
10 marks
- Q8**

Q8 C Code - Nested Loop

```
for(i=0;i<n;i++)  
for(j=0;j<n;j++)  
printf("**");
```

Your Answer

O(n2)

Save Answer

Q9 C Code – Binary Search Complexity (10 marks)

```
while(low<=high){  
mid=(low+high)/2;  
}
```

Answer: $O(\log n)$

Screenshot:

The screenshot displays the SIMATS Online Programming Lab interface. At the top, there is a dark blue header with the SIMATS ENGINEERING logo and the text "SIMATS Online Programming Lab" and "DS PROGRAMMING". Below the header, the page title is "CO1_AT1_C CODE OUTPUT PREDICTION". On the left side, there is a list of questions under the heading "QUESTIONS". The questions are numbered Q1 to Q8, each with a title and "10 marks". Q1 is "C Code – Array Traversal", Q2 is "C Code – Linear Search", Q3 is "C Code – Binary Search Mid", Q4 is "C Code – Pointer with Array", Q5 is "C Code – Array Update", Q6 is "C Code - Linked List Node", Q7 is "C Code - Time Complexity Loop", and Q8 is partially visible. The question Q3, "C Code – Binary Search Mid", is selected. On the right side, the question details are shown. It includes the question number "Q9" and the title "C Code - Binary Search Complexity". Below the title, the C code snippet is displayed:

```
while(low<=high){  
mid=(low+high)/2;  
}
```

 Under the code, there is a section titled "Your Answer" with a text input field containing the answer "O(logn)". At the bottom of the answer section, there is a blue button labeled "Save Answer".

Q10 C Code – Pointer Increment (10 marks)

```
#include<stdio.h>
int main(){
int a[]={5,10,15};
int *p=a;
p++;
printf("%d", *p);
}
```

Answer: 10

Screenshot:

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```
#include<stdio.h>
int main(){
int a[]={5,10,15};
int *p=a;
p++;
printf("%d",*p);
}
```

 Below the code, the "Your Answer" section shows the input "10". A blue button labeled "Save Answer" is at the bottom of the answer section.

QUESTIONS

- Q1** C Code – Array Traversal 10 marks ✓
- Q2** C Code – Linear Search 10 marks ✓
- Q3** C Code – Binary Search Mid 10 marks ✓
- Q4** C Code – Pointer with Array 10 marks ✓
- Q5** C Code – Array Update 10 marks ✓
- Q6** C Code - Linked List Node 10 marks ✓
- Q7** C Code - Time Complexity Loop 10 marks ✓
- Q8** ✓

Q10 C Code - Pointer Increment

```
#include<stdio.h>
int main(){
int a[]={5,10,15};
int *p=a;
p++;
printf("%d",*p);
}
```

Your Answer

10

Save Answer